Sound Is A Pressure Wave

SYNOPSIS

Notes 1 to 8 built the wave in general and then trapped one between two ends. This note asks what a *sound* wave is, and the answer is that it is a pressure wave. The air does not travel across the room with the sound — it crowds together and spreads apart in place, and what travels is the pattern. Everything else here follows from that: how fast the pattern moves, and what the medium has to do with it.

BEFORE YOU READ ON — PREDICT

Clap once in an empty room. **(1)** Does any air make the trip from your hands to the far wall? **(2)** At the instant the air at some spot is squeezed hardest, is that same air also moving fastest? **(3)** Sound runs through steel about fifteen times faster than through air. Is that because steel is *heavier*, or because it is *stiffer*?

PERFORMANCE EXPECTATIONS

1. **Describe** a sound wave as a longitudinal pressure variation about P_0, and define compression and rarefaction in terms of pressure.
2. **State** that displacement and pressure run a quarter wavelength out of step, and say which node goes with which antinode.
3. **Explain** the speed of sound as a stiffness over an inertia, and connect $\sqrt{(B/\rho)}$ to $\sqrt{(F_T/\mu)}$.
4. **Use** $v = 331.5 + 0.610T$ to find a speed from a temperature, and a temperature from a speed.
5. **Compute** a wavelength from a frequency, and check the answer against the size of a real object.

1. Compressions and rarefactions

A sound wave is a **longitudinal** wave in a material: the medium vibrates along the line the wave travels, not across it. Where the molecules crowd, the pressure sits above P_0; where they spread, below. One wavelength is the distance from one compression to the next.



Figure 1. Draw the molecules at one instant, then the pressure underneath. Mark one compression and one rarefaction, and bracket one wavelength.

1. Define a compression and a rarefaction in terms of *pressure* rather than crowding.

2. The molecule picture and the pressure graph are the same wave drawn twice. Which does a microphone measure, and why does that matter for the mathematics?

2. Displacement and pressure are not in step

This is the idea worth the most in this note. Two things vary along the wave: how far the molecules swing, and what the pressure does. They are a quarter of a wavelength apart.

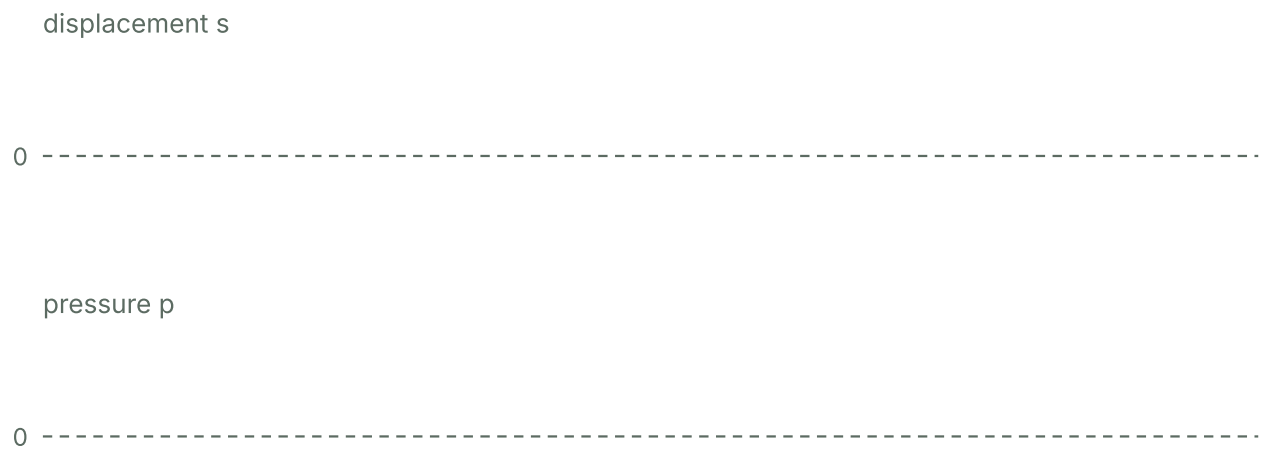


Figure 2. Draw the displacement on the upper axis and the pressure on the lower one. Mark a displacement node, and bracket the offset between the two curves.

3. Finish the sentence both ways round: a displacement **node** is a pressure _____, and a displacement **antinode** is a pressure _____.

4. Why should that be? Think about what the molecules on either side of a displacement node are doing while the node itself stands still.

Where the quarter wavelength comes from

Squeeze a slab of air and the pressure change depends on how much the displacement is *changing across* the slab, not on the displacement itself:

$$p(x,t) = -B\,\partial s/\partial x$$

Put the travelling wave of Note 7 into it and read off what kind of function the pressure is.

3. What the speed depends on

Pump the air out of a jar with a bell ringing inside and the bell goes quiet: there is nothing left to compress. What about the medium sets the speed?

Medium	Speed (m/s)	Medium	Speed (m/s)
Air (0 °C)	331.5	Alcohol	1,213
Carbon dioxide	258	Water	1,435
Hydrogen	1,269	Oak, with grain	3,850
		Steel	5,130

5. Write down the wave speed on a string from Note 8, and name what each symbol measures about the string.

6. Read the table. Write the rough pattern in one line each for gases, liquids and solids. Then say what hydrogen is doing there.

THE PUNCHLINE

Every wave speed in this module has the same shape: **a stiffness over an inertia, under a square root**. On a string, tension over mass per length. In a fluid, the bulk modulus over the density. Stiff and light is fast; floppy and heavy is slow.

4. And on temperature

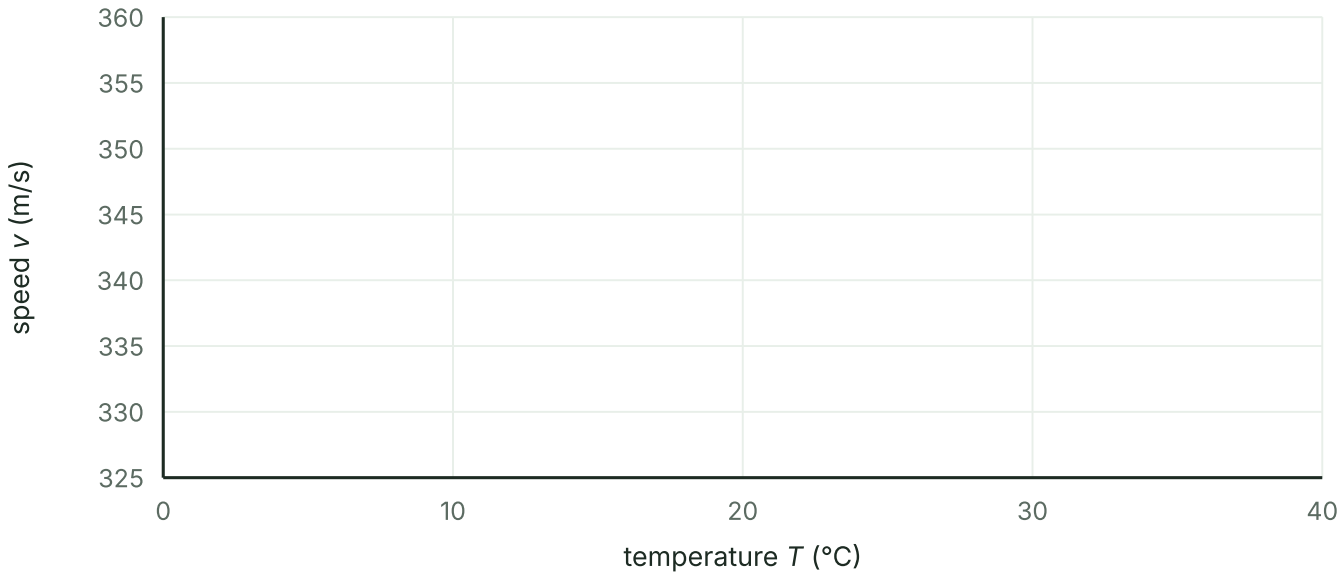


Figure 3. Plot the two points we measure, draw the line through them, and write its equation underneath.

$$v = _____ + _____ T$$

7. Room temperature is about 70 °F, about 20 °C. What is the speed of sound there?

8. You measure 338 m/s. How warm was the room?

9. You see lightning and hear thunder 10 s later. How far away did it strike?

5. One wavelength, to finish

With a speed in hand, $v = f\lambda$ from Note 2 turns any pitch into a length.

10. An A_3 on a cello is 220 Hz. Find its wavelength in room-temperature air, and check the answer against the size of a cello.

Where this sits. Note 8 trapped a wave between two ends. This note says what the wave is made of when the medium is air, and gives it a speed. Note 10 counts its energy, Note 11 follows that energy out into intensity and decibels, and Note 12 sets the source moving.

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